Research and Application of Multimodal Data Fusion in Medical Images

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***Abstract*-- The medical images produced by different medical detection equipment have different image modes, which have their own advantages and disadvantages. In order to improve the accuracy of medical diagnosis and make up for the limitations of single mode in information description, the different modes of the same disease information are fused to build diagnostic models, and the artificial intelligence technology and methods are used to organically combine multimodal medical data for clinical diagnosis and efficacy evaluation, Combining X-ray and CT scanning can improve the accuracy and reliability of the detection and diagnosis of lung diseases, help doctors evaluate the quality of the disease, reduce misdiagnosis and missed diagnosis, and also contribute to the formulation of treatment plans and disease monitoring.**

***Key words*-- medical images; Multimodal; Fusion of physical signs; diagnostic model**

I. Introduction

Modality refers to the way information is represented. In the field of data, multimodality is used to represent different forms of data, or different formats of the same form, typically representing text, pictures, audio, video, and mixed data. Multimodal data refers to data obtained from different domains or perspectives for the same description object, and each domain or perspective that describes these data is called a modality [1]. Multimodal data fusion mainly refers to the comprehensive processing of multimodal data using a computer, which is responsible for fusing information from various modes to perform target prediction [2].

The diversity of multimodal big data is one of the important characteristics that distinguish it from traditional big data. Multimodal big data is often composed of multiple data with different structural forms. The description of the same thing by these different modal data has both a certain degree of independence and strong relevance. Accurate and efficient extraction and processing of hidden fusion information from multimodal big data can help solve various current multimodal data analysis tasks.

Diagnosis and intervention of diseases play a positive role in ensuring and strengthening people's physical health. With the development of medical technology and various sensor technologies, and the widespread use of medical assisted diagnosis and sign detection, a large amount of diagnostic data has been generated, forming different modes of data for the same disease. Multimodal medical data has diversity, independence, and coupling. Accurate and efficient extraction and processing of fusion information hidden in multimodal medical data can help solve various current multimodal data analysis tasks, and has a positive effect on the diagnosis and treatment of diseases [3]. In recent years, machine learning has been widely used in medical image processing. Most of the main research focuses on building learning algorithms and models on multimedia images, while training sample sets in the same learning domain.

According to function, medical images can be divided into two categories: functional images and anatomical images. Functional images mainly include MRI and CT images to present morphological information of the human body, while anatomical images mainly include PET and SPECT images to describe functional and metabolic information of the human body. They provide multiple modal medical images of the same disease. Different modal images have different characteristics, and the display resolution of functional images is not strong, but they can provide metabolic information about organ functions that anatomical images cannot provide; Anatomic images can display anatomical information of organs well and have a strong resolution, which is

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incomparable to functional images.

At present, multimodal medical image fusion needs to solve many technical problems, and faces great challenges. Firstly, the fusion method has limitations and cannot be widely applied to multiple image fusion, and has not yet formed a complete set of fusion theories and unified fusion models; Secondly, due to the different imaging mechanisms of different modality medical images, different images have different formats, sizes, and qualities. Therefore, the preprocessing of medical images is strict and cumbersome, and it is very necessary to propose fusion methods with stable performance and good fusion results; Finally, after the fusion is completed, there is no unified medical image evaluation index, and doctors can only select images fused by different algorithms through their naked eyes and their own experience. In terms of objective evaluation standards, ready-made evaluation indicators are also used, and appropriate evaluation indicators are not formed based on the characteristics of medical images themselves. In view of the problems existing in the above medical image fusion technologies, it is urgent to develop new and effective fusion methods and appropriate evaluation indicators to solve the problems.

II. Steps of multimodal fusion

A. Multimodal information acquisition and extraction

Detect human medical data generated by devices using different detection methods, obtain a dataset, and extract information from it, mainly extracting structured, semi structured, or unstructured data from unstructured raw text. Multimodal information extraction is the main work of multimodal knowledge atlas research. For information extraction of different modes, single mode information extraction can be carried out first, followed by multimodal information processing. Medical image recognition mainly relies on algorithms such as scale invariant feature transformation and directional gradient histogram, and then combines machine learning algorithms for image recognition.

B. Multimodal representation learning

In order to facilitate the processing of extracted multimodal information, it is necessary to represent the input data. In the era of deep learning, the representation method used is to represent the input data as vectors, and then automatically extract features from the input data through the powerful modeling capabilities of deep neural networks. Knowledge representation learning is one of the basic tasks in the research and application of knowledge maps. Information can be transformed from single modal representation learning into numerical vectors that can be processed by machines or further abstracted into higher level feature vectors. Multimodal representation learning aims to reduce the distribution gap of modal information in the joint semantic subspace while maintaining the integrity of modal specific semantics.

C. Text Modal Representation

The core of text representation is to represent basic language units, then use neural networks to learn language models to extract text features, and finally use an output vector of the neural network as a text representation. In the early days, one hot representation words were used, and the representation of each word was an index of that word in the dictionary. However, this formal space consumption was large, and it could not model the semantic similarity between words, and there was a problem of data sparsity. The vector obtained by Mikolov using the neural network model is generally used as the word vector in the future. Neural networks for extracting text features mainly include simple feedforward neural networks, as well as cyclic neural networks that are good at sequence modeling, such as long-term and short-term memory networks and their variants. In recent years, Transformer based on self-attention mechanism and its variant BERT have better text modeling effects than RNN, gradually replacing RNN as the mainstream text feature extraction method.

D. Image modal representation

The convolutional neural network developed based on multi-layer neural networks is a deep learning method specifically designed for images, which has achieved

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excellent results in image processing. With the development of deep learning technology and the improvement of computer processing capabilities, multimedia data such as images can be analyzed using the same deep learning framework as text, which provides convenience for multimodal research [4].

III.Research methods for multimodal data fusion:

Multimodal fusion has a very wide range of applications, most of which are accomplished using model-based methods. According to the method of data fusion, it can be divided into early fusion (i.e., based on feature level fusion) and late fusion (i.e., based on decision level fusion). And hybrid fusion.

Early fusion, also known as feature fusion, refers to a fusion method that is performed immediately after feature extraction of a modal. The advantage of feature fusion is that it can utilize the correlation between multiple features from different modes at an early stage, and is suitable for situations where modes are highly correlated. For example, early fusion is used when combining audio and video features of speech recognition.

Post fusion, also known as decision level fusion, refers to the fusion performed after each mode has decided (classification or regression). For post fusion, it is necessary to use corresponding models to train different modes, and then fuse the output results of these models. Post fusion is the process of extracting features from multiple independent and different sources, fusing the extracted feature vectors in a certain manner, and then making decisions based on this joint feature. This fusion method can be expressed as:

$$y = D(d(x1), d(x2), \ldots, d(xn)) \quad (1\text{-}1)$$

$$d(xi) = Mi(f(xi)) \quad (1\text{-}2)$$

"Where, $xxxxfxnxddxddxxddxDD$ represents the decision-making judgment made by the model MDD.". The commonly used comprehensive decision-making methods in late fusion include: ① Average fusion: averaging the confidence scores of the output of a multimodal model to make a final prediction Weighted Fusion: Calculate a comprehensive confidence score by weighting the output confidence of a multimodal model to make a final decision.

Early fusion is a comprehensive decision made after each model (or method) makes its own decisions (such as classification or regression). This approach makes full use of the potential of each single mode classifier, and its output can be explained well, but compared with the early fusion, it ignores the mutual information between modes. This fusion method can be expressed as:

$$y = D\ (F\ (x1, x2, \ldots, xn)) \quad (1\text{-}3)$$

Wherein, $xxxfxfxxfnnFF$ represents the feature fusion method, and DD represents the decision method made by the model based on the fusion features. Feature based fusion methods Feature fusion methods can be divided into three categories: ① algorithms based on Bayesian decision theory; ② algorithms based on sparse representation theory; ③ algorithms based on deep learning theory. Due to the different representations of different modes and the different perspectives of reflecting information about things, there are some phenomena of information redundancy and information complementarity between multimodal information. Compared to other current feature fusion methods based on deep learning theory, various modal features are learned from a large amount of data, so they have significant advantages in terms of information redundancy and complementarity.

Hybrid fusion: Hybrid fusion combines the advantages of early fusion and late fusion, but also makes the structure of the model complex and increases the difficulty of training. Due to the flexibility and diversity of the structure of deep learning models, hybrid fusion methods are more suitable for use.

IV. Application of multimodal data fusion

In recent years, lung cancer, as a malignant tumor of the lung, has the second incidence rate and the highest mortality among all cancer patients (the 5-year relative survival rate of diagnosed lung cancer patients is 19% in

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general). It is of great significance to carry out early identification and diagnosis of lung cancer. The diagnosis of lung tumors in modern medicine can generally be divided into two parts: analytical diagnosis and pathological diagnosis. Analytical diagnosis often refers to various imaging modalities (such as ultrasound, X-ray, CT, MRI, etc.) for diagnosis. Pathological diagnosis is mainly achieved by extracting physiological samples and observing the histological structure and cytopathic characteristics using a microscope. Pathological diagnosis is more objective and accurate, so it is called the "gold standard" for the diagnosis of lung tumors. However, pathological examinations often have a certain degree of trauma. In order to avoid trauma during the examination process, this article uses artificial intelligence technology and methods to organically combine multimodal medical data for clinical diagnosis and efficacy evaluation. Combining X-ray and CT scanning can improve the accuracy and reliability of detection and diagnosis of lung diseases, and contribute to the formulation of treatment plans and disease monitoring.

This paper focuses on multimodal medical data fusion, and combines knowledge driven and data driven intelligent algorithms to solve multimodal medical image intelligent analysis problems. A cross modal multi data feature analysis system is constructed for cross modal data, and a feature extraction model for multi modal information fusion is designed on this basis; Utilizing the prior knowledge of medical image diagnosis, integrating two intelligent algorithms, knowledge driven and data driven, and combining the advantages of good robustness and interpretability of knowledge driven algorithms with good adaptability and high accuracy of data driven algorithms, to achieve intelligent analysis of multiple data medical images; And apply the research to the clinical diagnosis of lung disease.

A. Knowledge driven intelligent analysis of medical images.

Aiming at the fusion of prior knowledge and deep neural networks in medicine and imaging, discrete wavelet transform (DWT) is proposed to decompose the data into low-frequency components and high-frequency components, from which deep level features and high-frequency detail features are extracted. Finally, inverse discrete wavelet transform (IDWT) is used to fuse deep level features and high-frequency detail features to achieve the fusion of imaging knowledge and deep learning network design. Using operators such as Gabor, GSF, and SIFT to extract image features, guide and supervise the learning of deep neural networks at different locations, accelerate network learning and optimization, and achieve the fusion of imaging knowledge and network learning. According to medical diagnostic rules, personalized medical knowledge driven feature descriptors are designed for specific objects to extract image features with distinctive medical significance, which complement the features extracted by deep learning networks, and achieve the fusion of medical knowledge and network features.

B. Intelligent analysis of medical images based on multimodal data fusion.

Aiming at the problem of multimodal heterogeneous data fusion, this project will design a knowledge driven deep neural network for multimodal heterogeneous data fusion based on actual medical data and the clinician's diagnostic inference classification process. The network integrates medical knowledge driven description features and image knowledge driven deep learning features for multimodal data to form accurate inference classification results. Different types of features meet the constraints of feature differentiation and result homogeneity, forming a more powerful and interpretable deep network through collaborative interaction, thereby achieving accurate classification and diagnosis.

C. The application of knowledge driven multimodal medical image intelligent analysis methods in disease diagnosis.

Aiming at the application of knowledge driven multimodal medical image intelligent analysis methods, this article takes the diagnosis of lung diseases as the application direction to verify the effectiveness, practicality, and universality of knowledge driven

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multimodal medical image intelligent analysis methods. For lung CT images and X-ray films, design features consistent with medical knowledge to describe the lungs; For CT images, a knowledge driven multimodal data fusion deep neural network that conforms to the medical analysis and diagnosis process is designed. Finally, the features extracted from multiple modal data are integrated to make disease diagnosis, treatment recommendations, and prognosis estimates for patients. The design principle is to fully utilize medical knowledge to extract meaningful features, design a multimodal data fusion network that conforms to medical diagnostic reasoning, and incorporate imaging prior knowledge into network design and learning [5].

V. Summary

This paper designs a deep learning neural network that can achieve multimodal data input, knowledge driven feature extraction, and knowledge driven and data driven fusion. Based on this, a unified feature fusion learning theory and algorithm framework is developed, which integrates knowledge driven, feature extraction, and feature fusion. It fully integrates medical and imaging knowledge into all aspects of deep learning network design and learning, and designs a multimodal data fusion network that conforms to medical diagnostic reasoning, providing accurate clinical assistance to doctors. In terms of deep learning methods, a knowledge driven wavelet convolution network is designed; In the aspect of feature extraction, we designed a feature descriptor based on the criteria of clinical and imaging doctors' diagnosis, and a deep neural network guided by knowledge driven feature loss; In the aspect of feature fusion, a deep neural network for multi-source image and multi-dimensional feature fusion combining multiple fusion methods such as front-end, back-end, and full fusion is designed to achieve knowledge-based multimodal medical image intelligent analysis. Through the above method, high-precision diagnosis of multimodal medical images using artificial intelligence algorithms in multiple disease diagnosis is realized.

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